Supervised learning
k-Nearest Neighbors

Artificial Intelligence
CSC 480
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Adapted with permission from
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Types of learning

- **Supervised learning**
  Learning to predict or classify labels based on labeled input data

- **Unsupervised learning**
  Finding patterns in unlabeled data

- **Reinforcement learning**
  Learning well-performing behavior from state observations and rewards

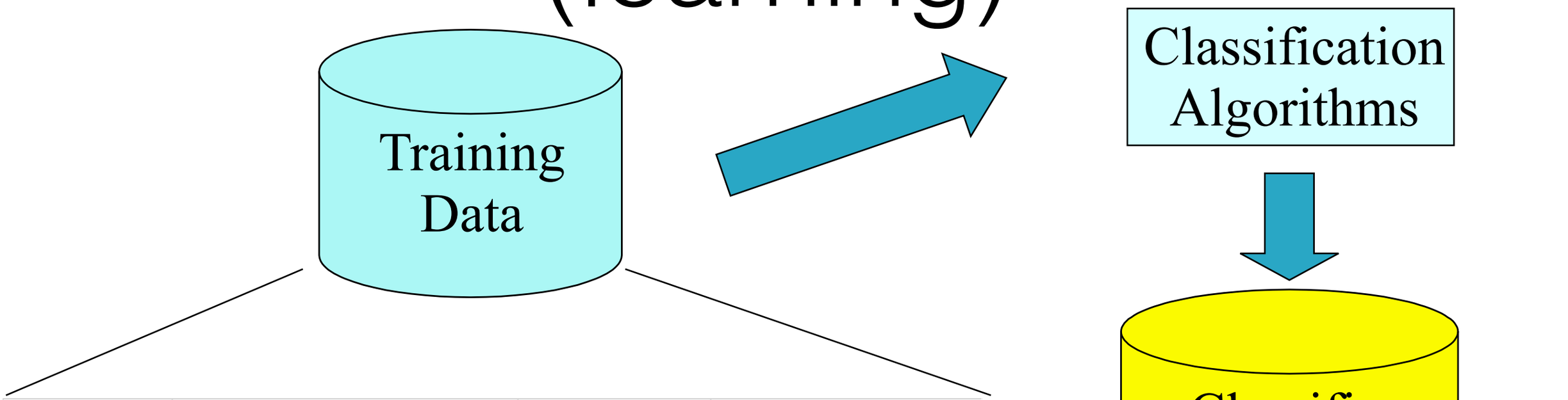
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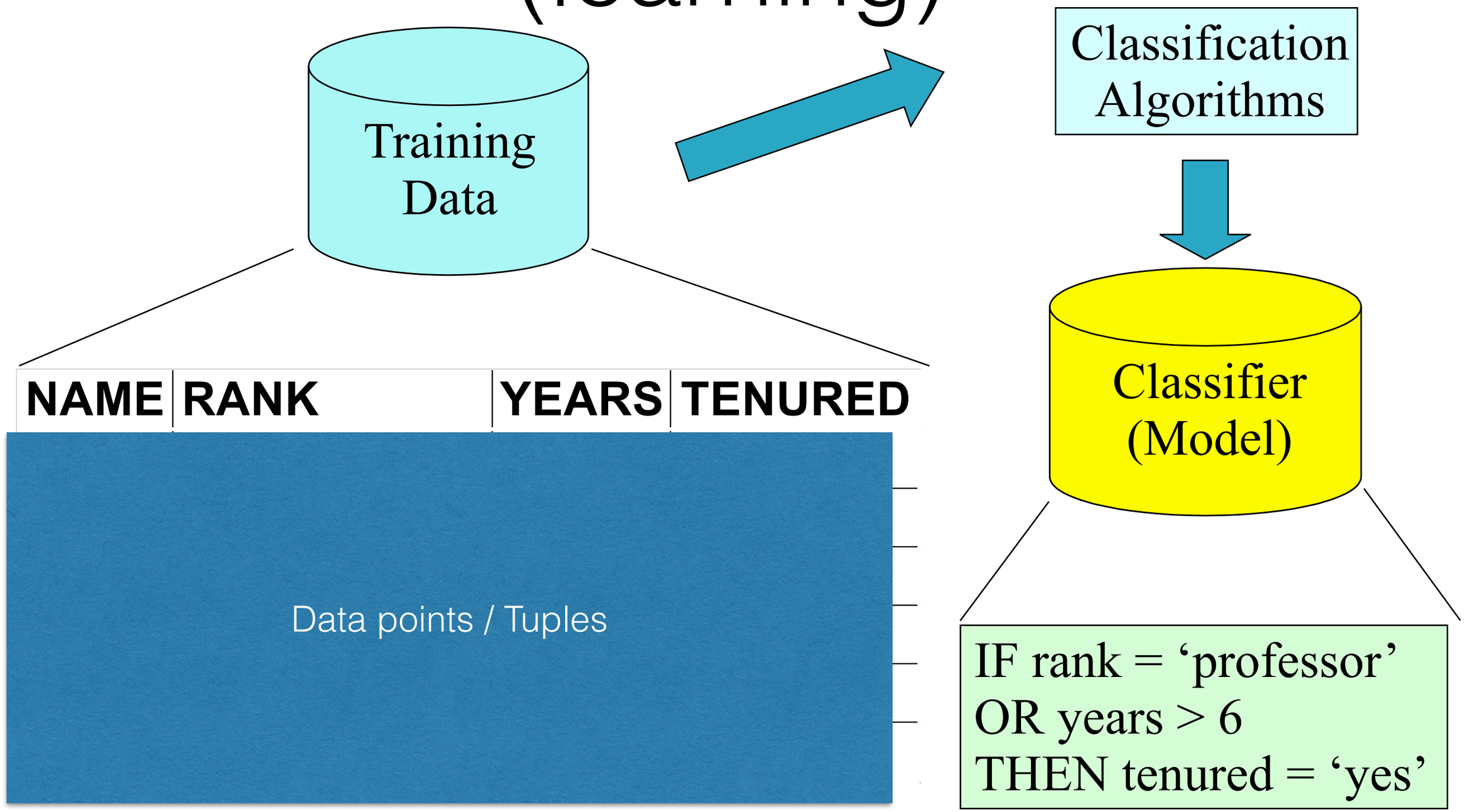
Model construction (learning)



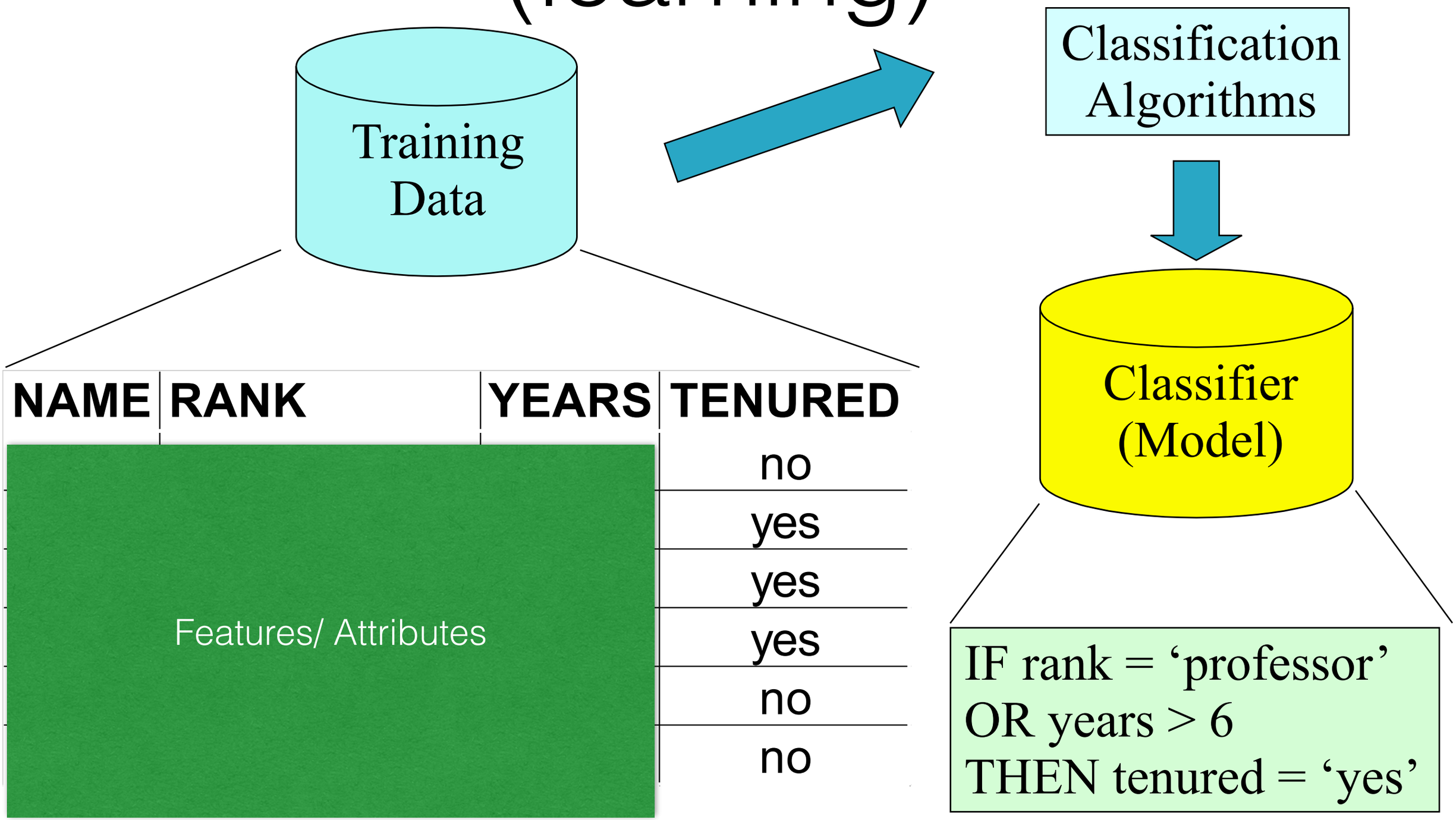
NAME	RANK	YEARS	TENURED
Mike	Assistant Prof	3	no
Mary	Assistant Prof	7	yes
Bill	Professor	2	yes
Jim	Associate Prof	7	yes
Dave	Assistant Prof	6	no
Anne	Associate Prof	3	no

IF rank = ‘professor’
OR years > 6
THEN tenured = ‘yes’

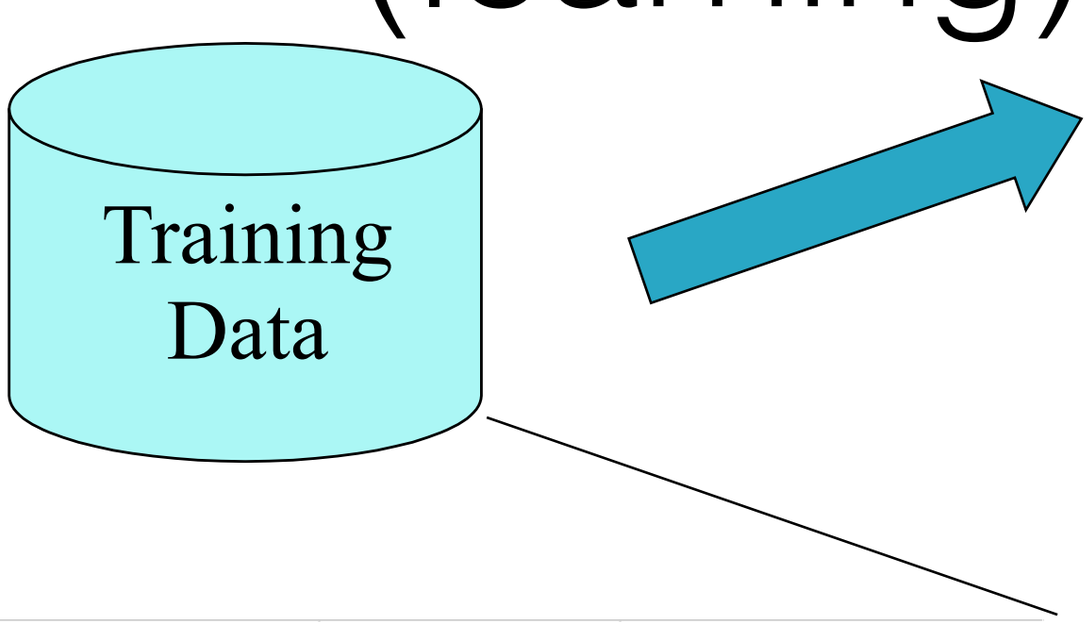
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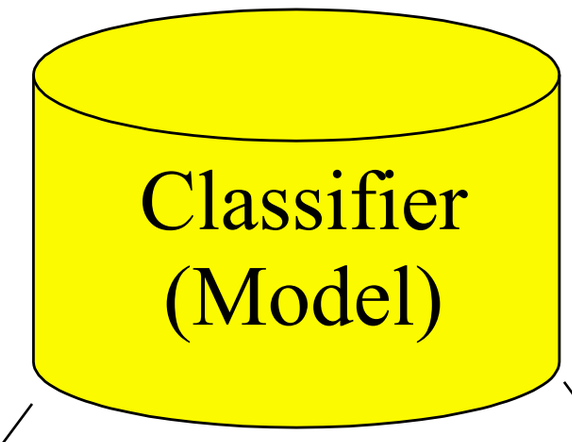


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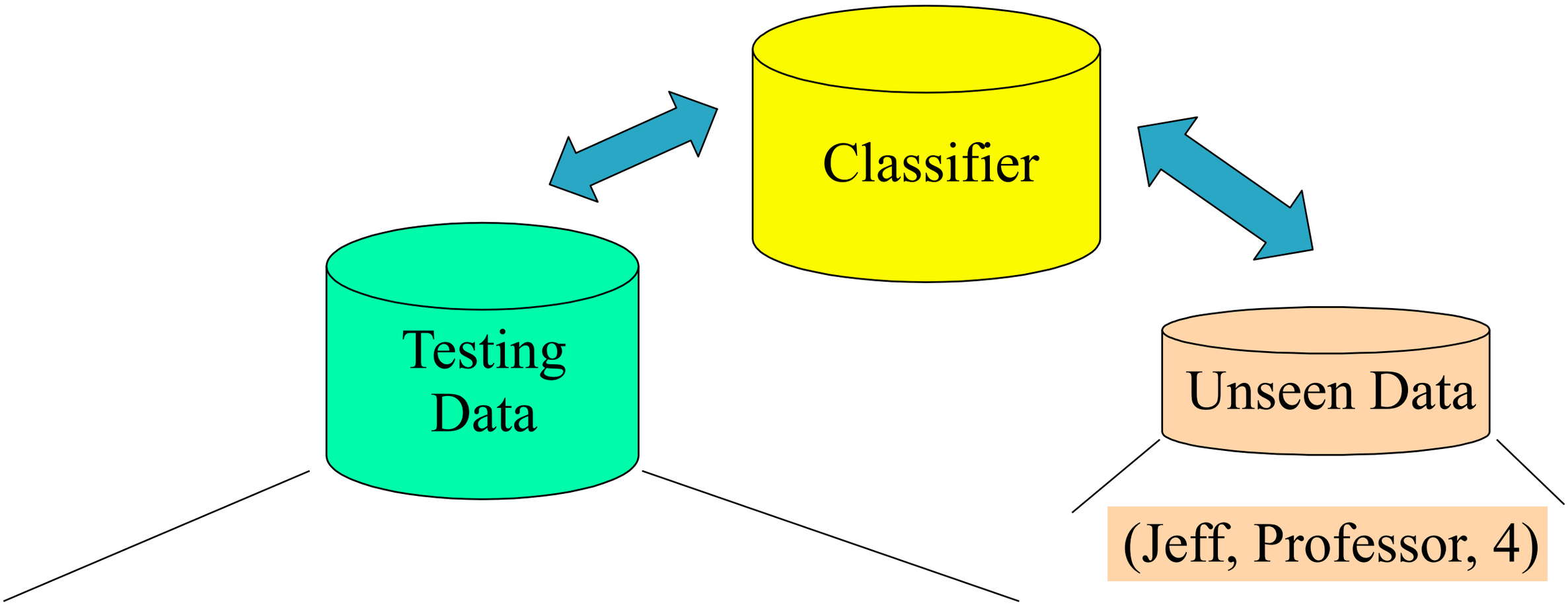


NAME	RANK	YEARS	TENURED
Mike	Assistant Prof	3	
Mary	Assistant Prof	7	
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Labels / Targets

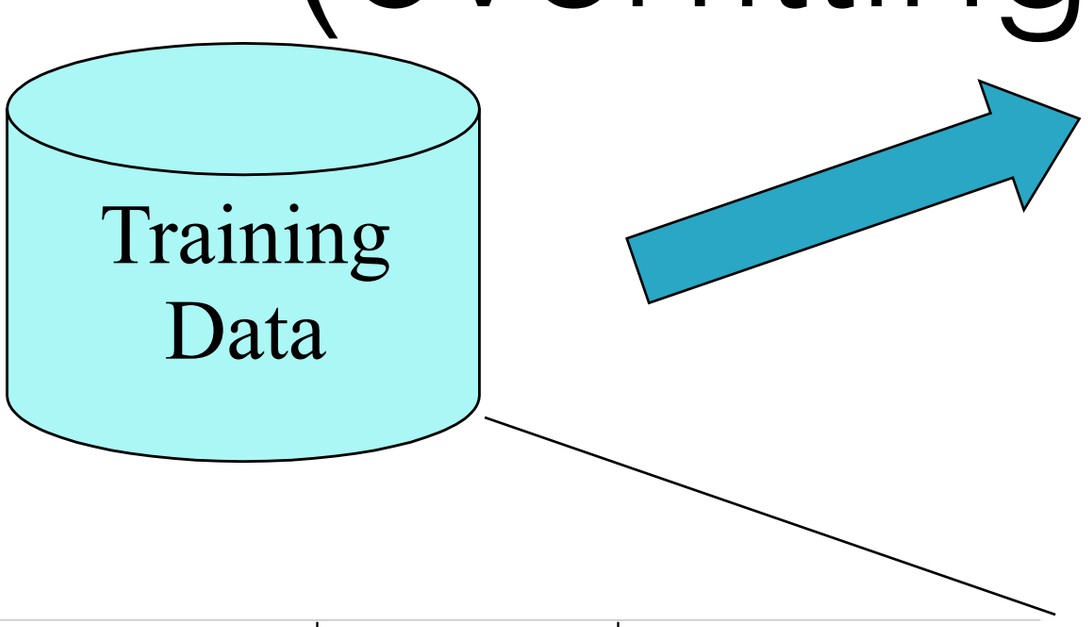
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OR years > 6
THEN tenured = ‘yes’

Using the model (inference)

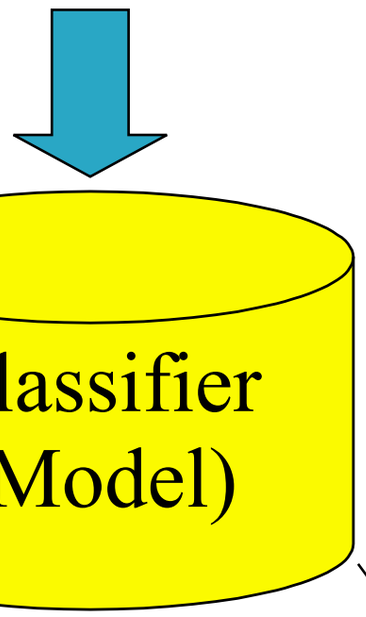


NAME	RANK	YEARS	TENURED
Tom	Assistant Prof	2	no
Merlisa	Associate Prof	7	no
George	Professor	5	yes
Joseph	Assistant Prof	7	yes

Model construction (overfitting)







NAME	RANK	YEARS	TENURED
Mike	Assistant Prof	3	no
Mary	Assistant Prof	7	yes
Bill	Professor	2	yes
Jim	Associate Prof	7	yes
Dave	Assistant Prof	6	no
Anne	Associate Prof	3	no

IF Name = “Mary”, “Bill” or”Jim” THEN Tenured = “yes”

Types of Datasets

- We train on the training set…
- …test <u>while learning</u> on the validation set…
- …and test <u>after learning</u> on the testing set (should be held-out until **after** learning. No peeking!)

Classification vs prediction

- Classification: binary or nominal labels
    - Examples: cat or dog, pregnant or not, from which country, which type of road sign
- Prediction: continuous labels
    - Examples: future stock price, life expectancy, distance to obstacle
    - Prediction is also called regression

What's desirable?

- Accuracy
  classifier accuracy: predicting class label
  regression accuracy: guessing value of predicted attributes

- Time / cost / energy…
  …to construct the model (training cost)
  …to use the model (classification/prediction cost)

- Robustness: handling noise and missing values

- Interpretability: humans can understand the model

- Generality: model should work on novel examples

- Other measures, e.g., goodness of rules, such as decision tree size or compactness of classification rules

Lazy vs Eager learning

- Lazy learning: Simply stores training data (or only minor processing) and waits until it is given a test tuple

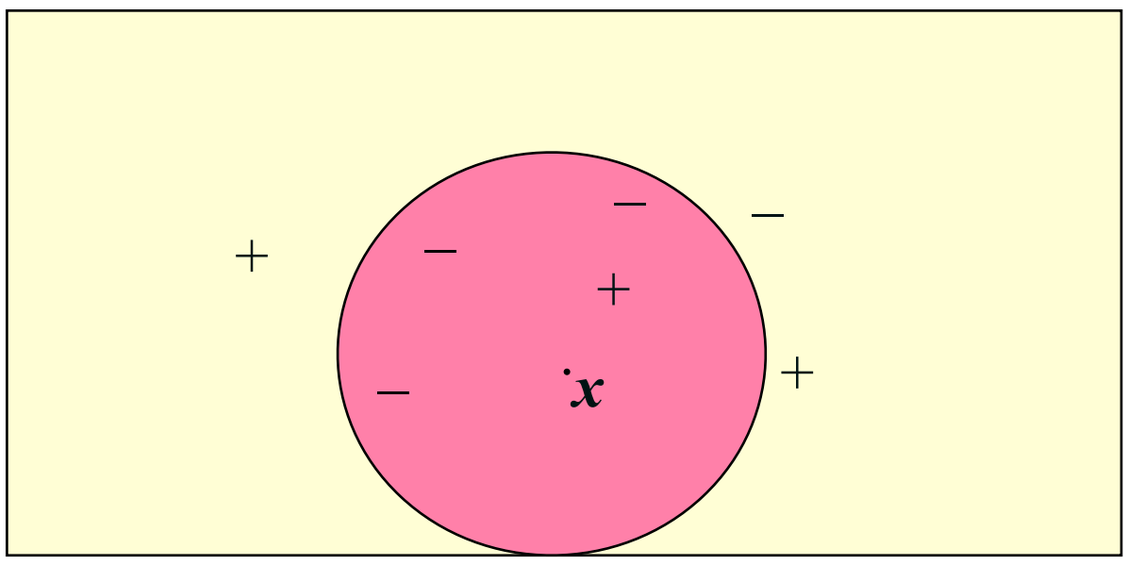
- Eager learning: Given a training set, constructs a classification model (smaller than the data) before receiving new data to classify

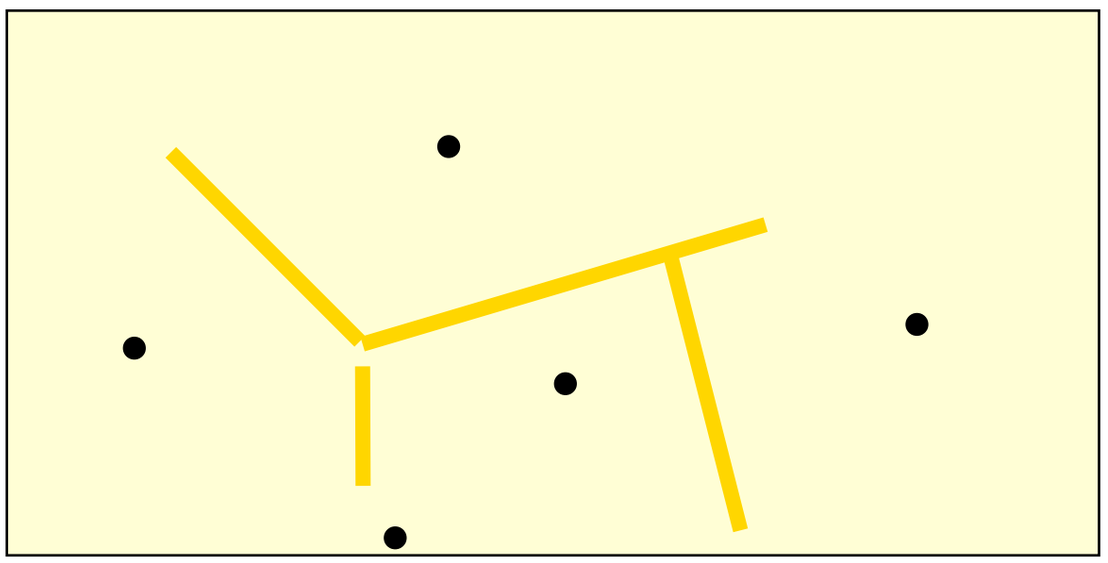
- Lazy: less time in training but more time in predicting

What's the simplest imaginable working classifier?

k-Nearest Neighbor Classification

- Simply look at the k instances in the training data which are closest to the instance you want to classify
- Choose the median/mean/mode of those values





k-Nearest Neighbor Classification

- All instances correspond to points in the n-D space
- The nearest neighbor is defined in terms of Euclidean distance, dist(X_1, X_2)
- Target function could be discrete- or real-valued
- Use the mode (most common value) for discrete targets, mean/median for real-valued targets
- Voronoi diagram: the decision surface induced by 1-NN for a typical set of training examples

k-Nearest Neighbor Classification

- Distance-weighted nearest neighbor algorithm: Weigh the contribution of each of the k neighbors according to their distance to the query X_q, and give greater weight to closer neighbors $w \equiv \frac{1}{d(x_q, x_i)^2}$

- Robust to noisy data by averaging k-nearest neighbors

- *Curse of dimensionality:* distance between neighbors could be dominated by irrelevant attributes

- To overcome it, stretch or shrink axes or eliminate the least relevant attributes

Distances

- Euclidean distance for continuous attributes

$$d(i,j)=\sqrt{(|x_{i1}-x_{j1}|^2+|x_{i2}-x_{j2}|^2+...+|x_{ip}-x_{jp}|^2)}$$

- Hamming distance for binary/nominal attributes: how many of the attributes differ